

Efficient SAR Vessel Detection for FPGA-Based On-Satellite Sensing

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Abstract

Rapid analysis of satellite imagery within minutes-to-hours of acquisition is increasingly vital for many remote sensing applications, and is an essential component for developing next-generation autonomous and distributed satellite systems. On-satellite machine learning (ML) has the potential for such rapid analysis, by overcoming latency associated with intermittent satellite connectivity to ground stations or relay satellites, but state-of-the-art models are often too large or power-hungry for on-board deployment. Vessel detection using Synthetic Aperture Radar (SAR) is a critical time-sensitive application in maritime security that exemplifies this challenge. SAR vessel detection has previously been demonstrated only by ML models that either are too large for satellite deployment, have not been developed for sufficiently low-power hardware, or have only been tested on small SAR datasets that do not sufficiently represent the difficulty of the real-world task. Here we systematically explore a suite of architectural adaptations to develop a novel YOLOv8 architecture optimized for this task and FPGA-based processing. We deploy our model on a Kria KV260 MPSoC, and show it can analyze a ~700 megapixel SAR image in less than a minute, within common satellite power constraints (<10W). Our model has detection and classification performance only ~2% and 3% lower than values from state-of-the-art GPU-based models on the largest and most diverse open SAR vessel dataset, xView3-SAR, despite being ~50 and ~2500 times more computationally efficient. This work represents

a key contribution towards on-satellite ML for time-critical SAR analysis, and more autonomous, scalable satellites.¹

CCS Concepts

• **Computing methodologies** → **Computer vision**; • **Hardware** → **Reconfigurable logic and FPGAs**; • **Applied computing** → **Earth and atmospheric sciences**.

Keywords

Edge AI, FPGA, Object Detection, On-Satellite Machine Learning, Synthetic Aperture Radar

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1 Introduction

Rapid analysis of low earth orbit satellite imagery within hours of data acquisition is vital for a diverse range of remote sensing applications, from wildfire detection, to agricultural monitoring, to disaster response [6, 8, 65], and latency requirements for many applications are moving beyond near real-time (1-3 hours), to quasi real-time (<30 mins), with target latencies of <5 minutes in some cases [39]. Furthermore, these shortest latency requirements will be increasingly important due to growing interest in autonomous and distributed satellite capabilities [4], for example “tip-and-cue” processes [15] (where one satellite uses initial analysis to request further imagery from another satellite). Such autonomous tasking and inter-satellite coordination capabilities have been identified as critical for next-generation Earth observation systems [69], with small test systems already in orbit (e.g. NASA’s Starling satellite swarm [22]). But achieving such rapid analysis is challenging, due to 1) ever-increasing dataset sizes of large-area, high-resolution

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¹Code: <https://github.com/alan-turing-institute/sar-vessel-detection-fpga>